Diffusion-RSCC: Diffusion Probabilistic Model for Change Captioning in Remote Sensing Images

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***Abstract*— Remote sensing image change captioning (RSICC) aims at generating human-like language to describe the semantic changes between bitemporal remote sensing image (RSI) pairs. It provides valuable insight into environmental dynamics and land management. Unlike conventional change captioning (CC) tasks, RSICC involves not only retrieving relevant information across different modalities and generating fluent captions but also mitigating the impact of pixel-level differences on the terrain change localization. Pixel-level discrepancies over a long time span decrease caption accuracy. To address these problems, we propose a probabilistic diffusion-based model that leverages its remarkable generative capability to produce flexible captions. In the training phase, we construct a condition denoiser to efficiently map the real caption distribution to a standard Gaussian distribution. This denoiser incorporates cross-mode fusion (CMF) and stacking self-attention (SSA) modules to enhance cross-modal alignment and reduce pixel interference, thereby improving caption accuracy. In the training phase, the condition denoiser provides a new strategy for mean value estimation and helps to generate captions step by step. Extensive experiments on the LEVIR-CC dataset and DUBAI-CC dataset demonstrate the effectiveness of our Diffusion-RSCC and each of its individual components. The quantitative results showcase superior performance over existing methods across both traditional and newly introduced metrics. The code is available at: https://github.com/Fay-Y/Diffusion-RSCC.**

***Index Terms*— Attention mechanism, change captioning (CC), diffusion model, remote sensing.**

I. Introduction

IN RECENT years, tasks related to multimodal data processing [1], [2], [3] in the field of remote sensing have attracted widespread attention. Among these tasks, change captioning (CC) [4], [5], [6] for remote sensing images (RSIs) has gradually emerged as a popular theme. Remote sensing image CC (RSICC) aims at generating human-like language sentences to describe the semantic change between bitemporal RSI pairs. It can provide accurate and insightful change captions, offering valuable perspectives into the dynamic nature of land cover [7].

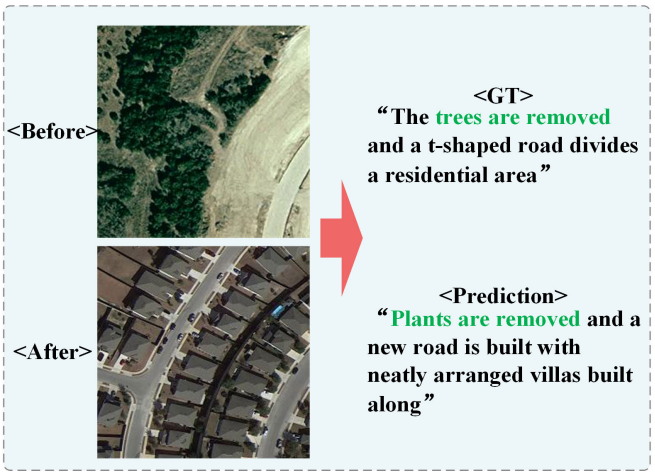


Fig. 1. Example from the LEVIR-CC dataset and the comparison between Diffusion-RSCC model prediction and the ground-truth (GT) caption. It demonstrates the ability of Diffusion-RSCC to describe the changes between the bitemporal image pairs.

Before exploring RSICC, it is important to understand the groundwork established by remote sensing change detection (RSCD) [8], [9]. RSCD tasks distinguish differences from similar images acquired in the same geographical location at different times. Models in this field have already achieved high accuracy in identifying natural changes from unrelated disturbances in bitemporal images. BGSINet-CD [65] introduces a graph semantic interaction module to suppress task-irrelevant information. SEIFNet [7] captures global and local information from bitemporal feature maps under semantic information for better feature representation. However, RSCD still has its limitations in providing comprehensive interpretation. Evolving from RSCD, RSICC transforms pixel-level differences into captions, making it convenient for humans to utilize RSI data.

Chouaf et al. [10] initiated CC in remote sensing, while Liu et al. [6] introduced the LEVIR-CC dataset, consisting of 10 077 pairs of bitemporal RS images and 50 385 sentences describing changes. Since its release, this dataset has become a vital resource for RSICC, enabling studies to address the pixel-difference challenge in RS datasets with extensive

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temporal spans [6]. Different from conventional CC tasks, RSICC involves not only retrieving relevant information across different modalities [11] but also mitigating the impact of pixel-level [12] differences on terrain change localization. A primary challenge in RSICC lies in distinguishing semantic changes from spurious changes, potentially compromising caption accuracy. Liu et al. [11], [12] addressed the challenge by decoupling the task into two stages: change detection and CC. By improving the change localization accuracy through the change detection phase, models were able to enhance the overall precision. Furthermore, RSICC faces a dearth of high-quality datasets, as images are derived from real satellite imagery [13], and textual annotations require manual effort [6]. Limited image–text pairs pose a significant challenge for learning an appropriate mapping from changes to natural language. In addition, autoregressive decoders [10], [15] fail to escape the fixed structures and style of sample sentences, restricting their ability to generate flexible and authentic captions that truly align with visual features. To enhance the CC algorithms performance and address data insufficiency, the study must address two crucial issues: 1) efficiently aligning the distributions of multitemporal and multimodal data to capture changes across time and modalities and 2) modifying the caption decoder to foster more flexible and accurate captions that are consistently aligned with the visual content.

Recently, denoising diffusion probabilistic models (DDPMs) [14] have shown incredible capabilities as generative models, which are able to recreate the true sample from a Gaussian noise input. They have showcased remarkable success in both continuous visual data [14], [15], [16] and discrete language data domains [17], [18], [19], exhibiting the capability to produce high-quality samples. These models facilitate a deep understanding of pixel connections, word connections, and image–text connections in multimodal tasks [20], [21]. Specifically, diffusion models are a class of likelihood-based models, where a Markov chain of diffusion steps is designed to slowly add random noise to data and then learn to reverse the diffusion process to construct desired data samples from the noise. Given the inherent randomness, diffusion models are capable of capturing the diversity and uncertainty in data during the training process. This randomness not only assists in reducing overfitting for generalization capabilities but also enables them to better accommodate the complexity and variability of real-world data. Considering the desirable properties of diffusion models, such as a stationary training objective, and easy scalability, we aim to bring the benefits to the field of CC and overcome the abovementioned challenges.

In this article, we propose a novel probabilistic diffusion model to capture semantic change in bitemporal RSIs. The proposed framework aims to learn the mapping from the standard Gaussian distribution to the distribution of real captions through condition denoiser. As for the testing phase, the model can incrementally denoise the standard Gaussian noise into vectors corresponding to words. The gradual denoising steps help produce hierarchical continuous latent representations and generate change captions under image control. Specifically, the proposed condition denoiser is composed of five main blocks: 1) pretrained CNN to extract features from the bitemporal image pairs; 2) time embedding block; 3) position embedding block to incorporate the temporal and spatial information; 4) cross-mode fusion (CMF) module to integrate the visual features and embedded sentences with spatiotemporal precision; and 5) stacking self-attention (SSA) module to iteratively refine the mutual attention score between cross-modal data for conditional mean value estimation. The condition denoiser, especially the CMF and SSA module, helps the diffusion model to present semantic changes with well-aligned input mapping. Based on the learning principle and data integration across modalities, Diffusion-RSCC can comprehend contextual information and generate semantically accurate and fluent descriptions, as shown in Fig. 1. Through this work, we significantly advance the field of RSICC with a nonautoregressive framework.

To sum up, our contributions consists of twofold that are as follows.

1) We present Diffusion-RSCC, a pioneering network that integrates a generative diffusion model into the CC field. This integration leverages the inherent randomness in diffusion models to enhance the complexity and variability of the generated captions while effectively reducing overfitting. Experiments demonstrate that our method achieves superior performance compared with existing approaches.
2) A novel condition denoiser with CMF and SSA modules is proposed. This denoiser is designed to effectively learn the cross-modal data distribution between image pairs and their corresponding change captions. It provides a new strategy for conditional mean value estimation during the reverse process. Ablation studies confirm the individual effectiveness of these modules.

The rest of this article is organized as follows. Section II introduces the works related to this article. Section III describes the methodology and introduces the model Diffusion-RSCC. Section IV shows the experimental results and ablation studies, and finally, Section V concludes this article.

II. Related Works

A. Diffusion Process

Diffusion probabilistic models emerge as formidable contenders alongside convolutional neural networks (CNNs) and attention mechanisms in computer vision and natural language processing. They have showcased remarkable generative power in artificial intelligence-generated content.

Diffusion models work as parameterized Markov chains trained using variational inference to produce samples matching the data after finite time. It consists of a two-phased process: in the forward diffusion process, the model gradually adds noise to the data by sampling until the signal is destroyed while in the reverse process, the model uses a noise predictor, for example, U-Net [22], to denoise the destroyed data from the forward process to achieve complete representation restoration.

The generative capabilities of diffusion models have been leveraged for robust feature extraction and representation.

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Since DDPMs [14] first introduced diffusion into computer vision, diffusion models find diverse applications in visual tasks, such as video generation [23], [24], [25], image editing [26], [27], and image super-resolution [15], [28], [29]. In remote sensing, IDF-CR [30] consisted of a pixel space cloud removal (Pixel-CR) module and a latent space iterative noise diffusion (IND) network and exhibits strong generative capabilities to achieve component divide-and-conquer (CDC) cloud removal. EDiffSR [31] with a novel noise predictor EANet can restore visual-pleasant images on both simulated and real-world RSIs.

Diffusion models have been extended to numerous fields. ControlNet [32] introduced multiple auxiliary condition paths, while stable diffusion ensured a stable diffusion process by projecting diffusion and sampling into the latent space. Diffusion-LM [18] introduced the diffusion model into natural language processing and used the BERTEncoder [33] as the noise predictor. It successfully transmitted the diffusion models from continuous image data to discrete words and showed the potential for application in multimodal tasks. Glide [34] with a U-Net noise predictor explored diffusion models for the problem of text-conditional image synthesis and it can be fine-tuned to perform image inpainting, enabling powerful text-driven image editing. SCD-Net [21] coupled the continuous diffusion process with a new guided Transformer-based sequence learning in image captioning.

However, the utilization of diffusion models in remote sensing cross-modal tasks remains largely unexplored. Inspired by these outstanding efforts, we leverage diffusion models with a new attention-based noise predictor for more accurate semantics and richer fine-grained details in RSICC.

B. Remote Sensing Change Detection

RSCD aims at locating pixel-level changes in the multitemporal RS images and generating change maps. RSCD has evolved from traditional approaches to advanced deep learning methods. These traditional techniques laid the groundwork for current RSCD methodologies.

Change vector analysis (CVA) [35] method calculated change intensity and direction between pixels to distinguish change regions using thresholds. Principle component analysis (PCA) [36] in change detection helped figure out variance in multitemporal images, while multivariate alteration detection (MAD) [37] method was applied to detect the semantic changes of corresponding multitemporal image scenes. The advanced method Li–Strahler geometric-optical model [38] can be combined with a scaling-based approach to detect structural changes and retrieve properties.

With the advent of deep learning, RSCD has seen substantial improvements. Deep learning models, including CNNs and Transformers, have enhanced the performance of change detection by leveraging superior multitemporal feature learning capabilities. DDCNN [39] with a dense attention method used the spatial context information to capture the changed features of ground objects. The PA-Former [40] can combine CNN-based feature extraction with a Transformer encoder and decoder for refined change discrimination. Diffusion models also play an important role in RSCD. GAD algorithm [41] fit semantic segmentation predictions to the input images for weakly supervised change detection. SwiMDiff [42] integrated contrastive learning with a pixel-level diffusion model and has been rigorously tested on change detection.

C. Change Captioning

CC is an emerging task in the vision-language community that extends beyond conventional image or video captioning by describing the differences between bitemporal images. Unlike traditional captioning, which focuses on a single image or video frame, CC must handle the added complexity of identifying and articulating the changes between image pairs.

CC tasks early spring up in surveillance video datasets and diagnostic datasets other than RSIs. The pioneer work with well-aligned image pairs in surveillance video datasets was first proposed by Jhamtani and Berg-Kirkpatrick [43]. The initial effort established the encoder–decoder structure as the paradigm for subsequent studies. Accordingly, plenty of researchers have explored CC task and proposed various solutions to address realistic challenges, including the presence of distractors and aligning image features with textual semantics. Park et al. [44] introduced the CLEVER-Change dataset, incorporating both semantic and pseudochanges due to viewpoint shifts. The proposed DUDA by Park et al. [44] compensated for the influence of illumination variations and localized the changes by direct subtraction of bitemporal image pairs. SRDRL [45] introduced the part-of-speech (POS) of words for switching visual information. CC-Full [46] combined reinforcement learning with text-based image manipulation TIM to refine change localization and captioning.

Other works focus on explicitly modeling the features of change that contribute to the alignment of multimodality input. VAM [47] employed a viewpoint-agnostic matching encoder to summarize and remove common properties between images to infer changes. VACC [48] predicted change features and refined sentences using a cycle consistency module. R3Net [49] utilized a representation reconstruction network and a syntactic skeleton predictor to enhance semantic interactions between change localization and caption generation. MCCFormers-S and MCCFormers-D [50] leveraged a Transformers encoder–decoder structure to model interactions between images implicitly and caption generation. Recent advancements have further refined these approaches. VARD [51] disentangled viewpoint and pseudochanges for more accurate change localization. These diverse approaches make rapid progress to tackle the challenges posed by viewpoint changes and pseudochanges in CC.

However, the RSICC task faces more challenges of limited datasets and pseudopixel-difference distractors. RSICCformer [6] with the proposed LEVIR-CC dataset set the benchmark for RSICC. PromptCC [5] addressed the primary challenge by distinguishing visual semantics from spurious changes and decoupling the task into change detection and captioning stages for the GPT prompt. To enhance algorithm performance, works in RSICC should address data insufficiency by efficiently aligning the distributions of multitemporal and multimodal data and modifying the decoder to generate more flexible and accurate captions.

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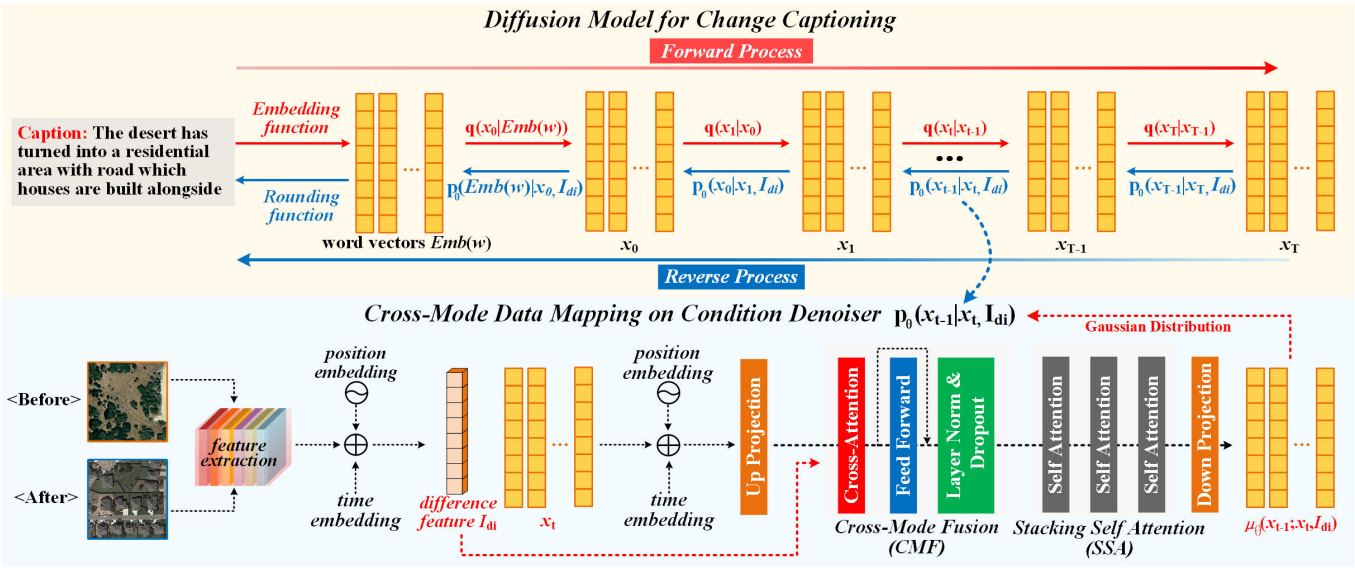


Fig. 2. Flowchart of our proposal. The diffusion framework consists of two processes: forward process to add noise to data until corruption and reverse steps to denoise with the condition denoiser. The condition denoiser consists of: 1) CNN-based feature extractor; 2) time embedding block; 3) position embedding block; 4) CMF module; and 5) SSA module.

Inspired by previous work, we explore the diffusion models to achieve better cross-modal alignment in RSICC.

III. Methodology

In this section, we propose a novel diffusion-based generative model for RSICC, as illustrated in Fig. 2. The proposed DDPMs are composed of two main parts: 1) the forward diffusion and the reverse denoising process and 2) the construction of condition denoiser for the CC task. The purpose of the forward diffusion process is to construct a mapping from the distribution of caption to standard Gaussian distribution, while the reverse process aims to denoise the prior standard Gaussian distribution back to the real data distribution using denoiser. It is worth mentioning that the construction of denoiser plays a vital role in our model, which establishes the mapping from image to language and decides the accuracy of the captions directly.

In Section III, we first introduce the forward and reverse steps of the diffusion models in Section III-A. Then, we will present the network architecture of the condition denoiser in Section III-B. The implementation details in the training and testing phase are presented in Section III-C.

A. Diffusion Process

According to [14] and [18], the diffusion model we utilized contains two processes: the forward diffusion process and the reverse denoised process.

1) Forward Process: CC task aims to sample w from a trainable language model $p(w|I_{\text{bef}}, I_{\text{aft}})$, where $w = [w_1, w_2, \ldots, w_n]$ is the sentence of discrete word with length n, I_{bef} and I_{aft} are the RSI pairs obtained over the same area with long time span. The embedding of the sentence w can be computed as: $\text{Emb}(w) = [\text{Emb}(w_1), \text{Emb}(w_2), \ldots, \text{Emb}(w_n)]$.

In forward process, noise is added following a Markov transition [18] from embedding representation $\text{Emb}(w)$ to x_0, following $q_\phi(x_0|w) = \mathcal{N}(\text{Emb}(w), (1-\alpha_0)\mathbf{I})$, where $\alpha_0 \in \mathcal{R}$ is a hyperparameter, ϕ is trainable parameters and $\mathbf{I} \in \mathcal{R}^{n\times d}$ is an identity matrix, as illustrated in Fig. 3. Then according to continuous diffusion step [14], the forward process can be formulated as follows:

$$q(x_t|x_{t-1}) := \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1-\alpha_t)\mathbf{I}) \tag{1}$$

where α_t is a hyperparameter. By the composition of the forward process from w to x_t, the joint distribution under w following the Markov chain can be computed as follows:

$$q(x_{0:T}|w) = q(x_0|w)\prod_{t=1}^{T} q(x_t|x_{t-1}). \tag{2}$$

2) Reverse Process: First, the rounding step with trainable parameters gets the discrete words from the continuous embeddings

$$p_\theta(w \mid x_0) = \prod_{i=0}^{n} p_\theta(w_i \mid x_{0,i}) \tag{3}$$

where $x_0 = [x_{0,1}, \ldots, x_{0,i}, \ldots, x_{0,n}]$ and $p_\theta(w_i \mid x_{0,i})$ is a linear projection layer followed by SoftMax activation function. θ represents the trainable parameters in the reverse process

As shown in Fig. 4, the Gaussian transition in the reverse denoised process of any adjacent latent states on Markov chains is formulated as follows:

$$p_\theta(x_{t-1} \mid x_t, I_{\text{bef}}, I_{\text{aft}}) = \mathcal{N}(\mu_\theta(x_{t-1}; x_t, I_{\text{bef}}, I_{\text{aft}}), \Sigma(t)) \tag{4}$$

where I_{bef} and I_{aft} are the bitemporal image pair. μ_θ and $\Sigma(t)$ are the denoised mean and variance of $p_\theta(x_{t-1} \mid x_t, I_{\text{bef}}, I_{\text{aft}})$.

In Section III-B, we construct a condition denoiser $f_\theta(x_t, t, I_{\text{bef}}, I_{\text{aft}})$ that can help to successfully predict the denoised mean value of $p_\theta(x_{t-1} \mid x_t, I_{\text{bef}}, I_{\text{aft}})$.

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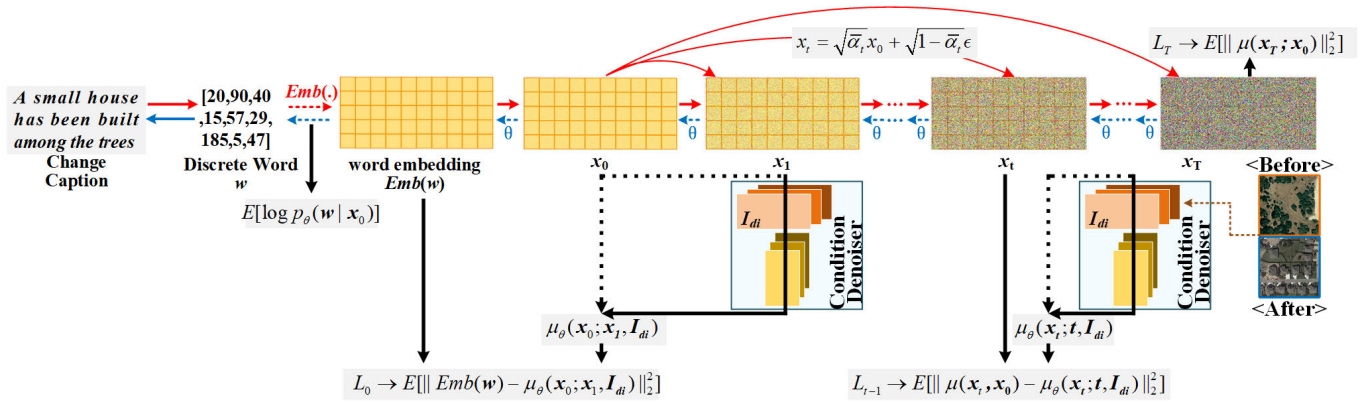


Fig. 3. Training process of the proposed diffusion model. The forward noising process is in red and the reverse denoising process is in black. The black arrows point out the four components of the loss function.

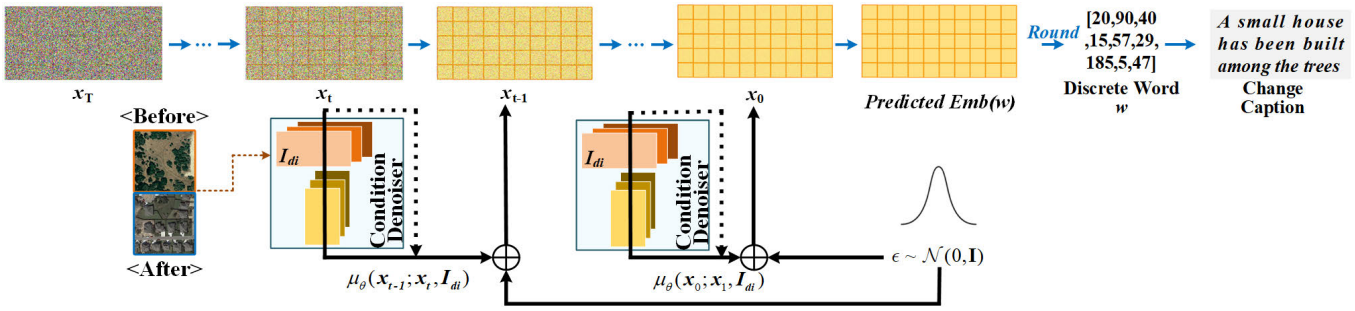


Fig. 4. Inference (testing) process of the proposed Diffusion-RSCC. At each timestep t, the condition denoiser estimates the mean value $\mu_\theta(x_t, I_{di}, t)$ of the posterior distribution.

3) Loss Function: The loss function aims at recreating a sampling distribution that is as close as possible to the prior data distribution, which can be achieved by maximizing the evidence lower bound of $p(w \mid I_{\text{bef}}, I_{\text{aft}})$

$$\begin{aligned}
&\log[p(w \mid I_{\text{bef}}, I_{\text{aft}})] \\
&\geq E_{q_\phi(-x_{0:T}|w)}\Bigg[-\log\frac{q(x_T \mid x_0)}{p_\theta(x_T)} \\
&\qquad - \sum_{t=2}^{T}\log\frac{q(x_{t-1} \mid x_0, x_t)}{p_\theta(x_{t-1} \mid x_t, I_{\text{bef}}, I_{\text{aft}})} \\
&\qquad - \log\frac{q_\phi(x_0 \mid w)}{p_\theta(x_0 \mid x_1, I_{\text{bef}}, I_{\text{aft}})} + \log p_\theta(w \mid x_0)\Bigg].
\end{aligned} \tag{5}$$

As follows [18], the loss function can be written as follows:

$$\mathcal{L} = \mathcal{L}_T + \Sigma_{t=2}^{T}\mathcal{L}_{t-1} + \mathcal{L}_0 - \mathcal{L}_{\text{round}}. \tag{6}$$

The four terms of (6) take the following form:

$$\mathcal{L}_T = D_{\text{KL}}(q(x_T \mid x_0)||p_\theta(x_T)) \tag{7}$$

where $q(x_T \mid x_0)$ and $p_\theta(x_T)$ are both Gaussian distributions. Considering that $p_\theta(x_T)$ is $\mathcal{N}(0, \mathbf{I})$ when T is sufficiently large, the KL divergence can be simplified

$$\mathcal{L}_T = E\left[||\mu(x_T; x_0) - 0||_2^2\right] = E\left[||\mu(x_T; x_0)||_2^2\right] \tag{8}$$

where μ is the mean value of data distribution in the forward process. Similarly, $\mathcal{L}_{t-1}$ and $\mathcal{L}_0$ are converted into the mean square error between the forward mean value and reverse mean value at corresponding step t

$$\mathcal{L}_{t-1} = E\left[||\mu(x_{t-1}; x_t, x_0) - \mu_\theta(x_{t-1}; x_t, I_{\text{bef}}, I_{\text{aft}})||_2^2\right] \tag{9}$$

$$\begin{aligned}
\mathcal{L}_0 &= E_{q_\phi(x_{0:T}|w)}\left[\log\frac{q_\phi(x_0 \mid w)}{p_\theta(x_0 \mid x_1, I_{\text{bef}}, I_{\text{aft}})}\right] \\
&= E\left[||\text{Emb}(w) - \mu_\theta(x_0; x_1, I_{\text{bef}}, I_{\text{aft}})||_2^2\right].
\end{aligned} \tag{10}$$

Then, the loss of rounding step can be recalculated as the cross-entropy form

$$\mathcal{L}_{\text{round}} = E_{q_\phi(x_{0:T}|w)}[\log p_\theta(w \mid x_0)]. \tag{11}$$

The parameters ϕ of embedding function $\text{Emb}(\cdot)$ and rounding function $p_\theta(w \mid x_0)$ can be trained in $\mathcal{L}_0, \mathcal{L}_{\text{round}}, \mathcal{L}_T$.

B. Cross-Mode Data Mapping on Condition Denoiser

In this section, we construct a condition denoiser network based on a cross-modal attention mechanism for the denoising process. The network architecture is shown in Fig. 5, and the detailed network architecture is as follows.

1) Feature Extraction: Inspired by the previous works [4], [5], [6], we encode each "before–after" image pair $I_{\text{bef}}, I_{\text{aft}}$ with ResNet101 [53] to extract the high level feature representations $\varphi(I_{\text{bef}}), \varphi(I_{\text{aft}}) \in \mathbb{R}^{C\times H\times W}$, where C, H, and W indicate the number of channels, height, and width, respectively. And then, the extracted features are flattened to $C \times HW$.

To capture the distinction of the bitemporal image pair, we obtain the residual map of the flattened features by

$$I_{di} = \varphi(I_{\text{bef}}) - \varphi(I_{\text{aft}}). \tag{12}$$

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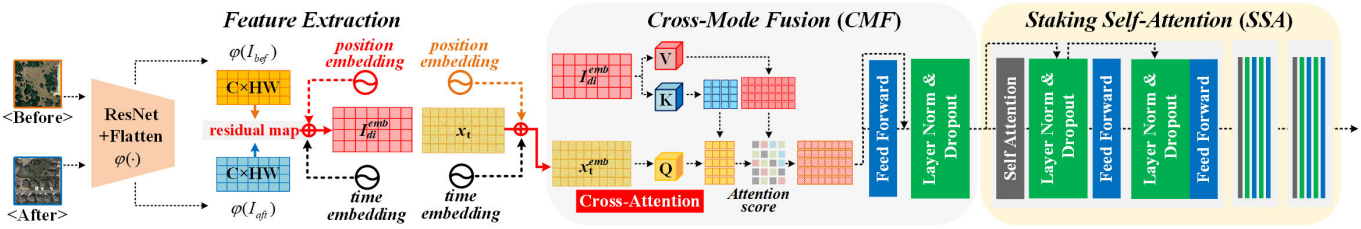


Fig. 5. Network architecture of the proposed condition denoiser: 1) feature extraction; 2) position embedding; 3) time embedding; 4) CMF; and 5) SSA.

Then, the residual maps are fed to the condition denoiser $f_\theta(x_t, t, I_{\text{bef}}, I_{\text{aft}}) = f_\theta(x_t, t, I_{di})$ and help to compute the mean value μ_θ of $p_\theta(x_{t-1} \mid x_t, I_{\text{bef}}, I_{\text{aft}}) = p_\theta(x_{t-1} \mid x_t, I_{di})$ in the reverse process.

2) Position Embedding: As shown in Fig. 5, the position embedding technique is introduced in both feature extraction I_{di} and latent representation of x_t. The main reason is that, we propose an attention-based network for the denoising step in the following section. The function used for position embedding are $P_x \in \mathbb{R}^{n \times d_{\text{model}}}$ and $P_I \in \mathbb{R}^{d_{\text{image}} \times d_{\text{model}}}$ which are trainable vectors with the same dimension of x_t and I_{di}, respectively.

3) Time Embedding: As shown in Fig. 5, time embedding is introduced in the latent representation of x_t since the proposed framework is diffusion-based [55]. Time embedding is devised on sine and cosine functions as shown in the following equation:

$$\begin{aligned} \text{TE}_{(t,2i)} &= \sin\left(\frac{t}{10\,000^{\frac{2i_t}{d_{\text{model}}}}}\right) \\ \text{TE}_{(t,2i+1)} &= \cos\left(\frac{t}{10\,000^{\frac{2i_t}{d_{\text{model}}}}}\right) \end{aligned} \tag{13}$$

where t is the timestep and i_t is the dimension.

4) Cross-Mode Fusion: CMF combines the cross attention and multihead attention to compute the mutual weights between semantic image features and textual sequence for well-aligned mapping. As shown in Fig. 5, CMF takes x_t^{emb}, I_{di}^{emb} as the input, where x_t^{emb} and I_{di}^{emb} are from the embedded representation x_t and I_{di}, which are concatenated with position embedding P_i and time embedding TE as follows:

$$\begin{aligned} x_t^{\text{emb}} &= x_t + P_x + \text{TE} \\ I_{di}^{\text{emb}} &= I_{di} + P_i + \text{TE}. \end{aligned} \tag{14}$$

As for the cross-multihead attention module, the query is from x_t^{emb}, while key and value come from the change feature I_{di}^{emb}. Also, the cross multihead attention $\text{CMHA}(\cdot,\cdot)$ in CMF module can be formulated as follows:

$$\begin{aligned} &\text{CMHA}(x_t^{\text{emb}}, I_{di}^{\text{emb}}) \\ &\quad = \text{Concat}(\text{head}_1, \ldots, \text{head}_h) \end{aligned} \tag{15}$$

$$\begin{aligned} &\text{head}_j \\ &\quad = \text{MultiheadAttention}\left(x_t^{\text{emb}} W_{hj}^Q, I_{di}^{\text{emb}} W_{hj}^K, I_{di}^{\text{emb}} W_{hj}^V\right) \end{aligned} \tag{16}$$

where $W_{hj}^Q \in \mathbb{R}^{d_q \times d_{\text{model}}}$, $W_{hj}^K \in \mathbb{R}^{d_k \times d_{\text{model}}}$, and $W_{hj}^V \in \mathbb{R}^{d_v \times d_{\text{model}}}$ are the parameter matrices of the jth head, and d_{model} is the hidden size.

To increase the nonlinearity of the network, additive attention computes a compatibility function using a feed-forward network which consists of two linear layers FC with an ReLU activation function. It can be formulated as follows:

$$\begin{aligned} x_t^{\text{proj}} &= \text{FC}\left(\text{ReLU}\left(\text{FC}\left(\text{CMHA}\left(x_t^{\text{emb}}, I_{di}^{\text{emb}}\right)\right)\right)\right) \\ x_t^{\text{fuc}} &= \text{LN}\left(x_t^C + \text{Dropout}\left(x_t^{\text{proj}}\right)\right) \end{aligned} \tag{17}$$

where LN denotes a layer norm and x_t^C is the output of $\text{CMHA}(\cdot,\cdot)$. Then, the fine-grained mapping output of CMF is fed to the following SSA module.

5) Stacking Self-Attention: Self-attention can recompute the representation of input x_t^{fus}, relating different positions of a single sequence. Through the SSA, the condition denoiser helps reduce x_t^{fus} noise and predicts a new mean value with structural coherence. The query, key, and value all derive from the CMF output x_t^{fus}

$$x_t^l = \text{Self-Attention}_l\left(x_t^{\text{fus}} W_l^Q, x_t^{\text{fus}} W_l^K, x_t^{\text{fus}} W_l^V\right) \tag{18}$$

where W_l^Q, W_l^K, and W_l^V are the trainable matrices of the lth self-attention layer.

After the self-attention layer (18), x_t^l is passed through a complementary attention block which takes the following terms:

$$\begin{aligned} x_t^{\text{res}} &= \text{LN}\left(\text{Dropout}\left(\text{FC}\left(x_t^l\right)\right) + x_t^l\right) \\ x_t^{\text{act}} &= \text{GeLU}\left(\text{FC}\left(x_t^{\text{res}}\right)\right) \\ f_\theta(x_t, t, I_{di}) &= \text{LN}\left(x_t^{\text{act}} + \text{Dropout}\left(\text{FC}\left(x_t^{\text{act}}\right)\right)\right). \end{aligned} \tag{19}$$

Our method can use the attention-based condition denoiser to reconstruct the mean value $\mu_\theta(x_{t-1}; x_t, I_{di})$ of the reverse process $p_\theta(x_{t-1} \mid x_t, I_{di})$ at step t and force the model to iteratively denoise for natural caption w generation under the image change feature I_{di}.

C. Training and Testing

In the training phase (Algorithm 1), as illustrated in Fig. 3, we iteratively optimize the condition denoiser by minimizing the loss function, ensuring that the model learns to generate accurate and semantically meaningful captions. In the testing phase (Algorithm 2), as illustrated in Fig. 4, given an input image pair $(I_{\text{bef}}, I_{\text{aft}})$, we utilize the trained condition denoiser to approximate the posterior distribution $p_\theta(x_{t-1}|x_t, I_{\text{bef}}, I_{\text{aft}}) = p_\theta(x_{t-1}|x_t, I_{di})$. Our method generates high-quality change captions w that accurately describe the changes on image feature I_{di}.

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Algorithm 1 Training Phase

Require: Total diffusion timestep T, caption w, image pair I_{bef} and I_{aft}, hyper-parameters $\alpha_0, \alpha_1, \ldots, \alpha_T$.

Initialize: ResNet101 network with flatten $\varphi(\cdot)$, word embedding function $Emb(\cdot)$, Condition Denoiser $f_\theta(\cdot)$ and rounding function $p_\theta(\cdot)$.
repeat
  Compute the image feature: $I_{di} = \varphi(I_{bef}) - \varphi(I_{aft})$
  Select randomly $t \sim \text{Uniform}(1, 2, \ldots, T)$
  Select randomly $\epsilon \sim \mathcal{N}(0, \mathbf{I})$
  Compute the start point: $x_0 = Emb(w) + \sqrt{1-\alpha_0}\epsilon$
  Compute current step at t: $x_t = \sqrt{\overline{\alpha}_t}x_0 + \sqrt{1-\overline{\alpha}_t}\epsilon$
  Predict denoised mean value: $\mu_\theta(x_{t-1}; x_t, I_{di}) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)f_\theta(x_t,t,I_{di})}{1-\bar{\alpha}_t}$
  According to (6)-(11), compute the loss function:
  if $t = 0$ **then**
    $\mathcal{L} = (\sqrt{\overline{\alpha}_T}x_0)^2 + (Emb(w) - \mu_\theta(x_t, t, I_{di}))^2$
      $-\log(p_\theta(w \mid x_0))$
  else
    $\mathcal{L} = (\sqrt{\overline{\alpha}_T}x_0)^2 + (x_0 - f_\theta(x_t, t, I_{di}))^2$
      $-\log(p_\theta(w \mid x_0))$
  end if
  Optimize θ by minimizing $\mathcal{L}$.
until convergence
return θ

Algorithm 2 Testing Phase

Require: diffusion timesteps T, image pair I_{bef}, I_{aft}, hyper-parameters $\alpha_0, \ldots, \alpha_T$, rounding function $p_\theta(\cdot)$.
Initialize: ResNet101 with flatten $\varphi(\cdot)$, Condition Denoiser $f_\theta(\cdot)$, rounding function $p_\theta(\cdot)$, $x_T \sim \mathcal{N}(0, \mathbf{I})$.
Compute the image feature: $I_{di} = \varphi(I_{bef}) - \varphi(I_{aft})$.
for $t = T, T-1, \ldots, 1$ **do**
  Sample $\epsilon_t \sim \mathcal{N}(0, \mathbf{I})$
  Sample x_{t-1} from $\mathcal{N}(\mu_\theta(x_{t-1}; x_t, I_{di}), \Sigma(t))$:
  $x_{t-1} = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})x_t + \sqrt{\bar{\alpha}_{t-1}}(1-\alpha_t)f_\theta(x_t,t,I_{di})}{1-\bar{\alpha}_t} + \frac{(1-\alpha_t)(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\epsilon_t$
end for
Select word w_i with the highest probability $p_\theta(w_i|x_0)$ for token x_0:

$$p_\theta(w|x_0) = \prod_{i=1}^{n} p_\theta(w_i|x_{0,i})$$

return $\mathbf{w} = [w_0, w_1, \ldots, w_n]$ as the changed caption prediction.

IV. Experiments

In this section, we describe the experimental results of our model and previous state-of-the-art methods through several experimental setups on the LEVIR-CC dataset and DUBAI-CC dataset. We also conduct ablation experiments on different substructures, including image feature extraction, mapping of image features and caption vectors, and self-attention blocks to demonstrate the effectiveness of each module.

A. Datasets

1) LEVIR-CC: It is a large-scale CC dataset proposed by Liu et al. [6]. It contains 10 077 image pairs (5038 changed image pairs and 5039 unchanged image pairs) from 20 different regions in TX, USA. The images cover a time span ranging from 5 to 15 years and are sized at 256 × 256 pixels with a resolution of 0.5 m/pixel. Within this dataset, each changed image pair exhibits one to two instances of alternation with five kinds of ground surface changes including buildings, roads, parking lots, vegetation, and water. Each image pair has five captions, 50 385 descriptive captions in total. The maximum caption sentence length is 39 words, with an average of 7.99 words. For training and testing, the dataset is partitioned into training, validation, and testing sets with 6815, 1333, and 1929 samples. Our vocabulary comprises the words that appear in the annotated captions, totaling 998 unique words. Notably, while previous works [5], [6] tend to remove words with a frequency below 5 to reduce data sparsity, we retain these low-frequency words to preserve the semantic integrity of the captions and ensure that the model can learn from a wider vocabulary distribution.

2) DUBAI-CC: It is a bitemporal CC dataset proposed by Hoxha et al. [66]. It consists of 500 image pairs from the Dubai area, captured by the Enhanced Thematic Mapper Plus (ETM+) sensor onboard Landsat 7 on May 19, 2000 and June 16, 2010. Each image is cropped into 50 × 50 pixel tiles, resulting in 500 tiles. The dataset is partitioned into training, validation, and testing sets, comprising 300, 50, and 150 tiles, respectively. These images are annotated with five captions per tile, yielding 2500 descriptive captions in total. The dataset focuses on changes in six categories: roads, houses, buildings, green areas, lakes, and islands. The maximum caption length is 23 words, with an average of 7.35 words. The vocabulary contains 266 unique words derived from the annotated descriptions. DUBAI-CC provides detailed annotations for urbanization changes over a ten-year period, offering valuable data for research in multitemporal remote sensing analysis. Notably, all DUBAI-CC images are resized to 256 × 256 pixels, aligning with the LEVIR-CC dataset.

B. Experimental Setup

1) Evaluation Metrics: The evaluation of the CC models is essential in assessing how well the generated sentences align with human judgments of the changes between image pairs. Traditional automatic evaluation metrics provide a measure of the accuracy of the generated captions based on annotated reference sentences. In this article, we utilize five commonly used metrics in CC tasks [44], [49], [56] to evaluate the accuracy of all methods: BLEU-N (where $N = 4$) [57], ROUGE-L [58], METEOR [59], CIDEr [60], and SPICE [61]. These metrics calculate the consistency between the predicted sentences and reference sentences. The candidates with higher scores indicate a greater degree of similarity and higher captioning accuracy. Specifically, we augment traditional metrics with additional indicators that are more suitable for evaluating captions with high variability: BARTScore [62] and MoverScore [67].

2) Implementation Details: We implemented all model training and evaluation on the Torch DL framework with a 24-GB NVIDIA RTX 4090 GPU. Following prior works, we use the ResNet101 network pretrained on ImageNet to extract image features. In the training phase, the model is trained for 564 epochs and 2000 diffusion steps with a batch size of 32. The total number of parameters is 58 423 318. We set the dimension of word embedding to 16 and built the vocabulary dictionaries with sizes of 998 and 266 for

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TABLE I

QUANTITATIVE COMPARISON OF DIFFUSION-RSCC WITH OTHER CC MODELS ON THE LEVER-CC DATASET. OUR PROPOSED MODEL OUTPERFORMS ALL BASELINES ON BLEU-4, METEOR, ROUGE-L, BARTSCORE, AND MOVERSCORE. THE NUMBERS ARE IN %. THE BEST RESULTS ARE IN BOLD

Method	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE	BARTScore	MoverScore
DUDA	57.6	37.7	71.1	**126.6**	29.3	-419.6	30.9
RSCCformer	49.6	28.5	59.3	97.4	20.9	-441.1	24.9
MCCFormers-S	46.2	27.2	58	94.3	20.3	-443.6	23.9
MCCFormers-D	31.5	24.2	52.1	77.4	22.2	-450.	24.5
VARD	43.2	33.5	66.6	113.7	**33.9**	-437.3	30.2
Diffusion-RSCC (Ours)	**60.9**	**37.8**	**71.5**	125.6	26.7	**-417.7**	**32.3**

TABLE II

QUANTITATIVE COMPARISON OF DIFFUSION-RSCC WITH OTHER CC MODELS ON THE DUBAI-CC DATASET. OUR PROPOSED MODEL OUTPERFORMS ALL BASELINES ACROSS ALL METRICS. THE NUMBERS ARE IN %. THE BEST RESULTS ARE IN BOLD

Method	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE	BARTScore	MoverScore
DUDA	17.9	21.7	39.4	24.9	16.3	-480.5	25.2
RSICCformer	28.9	21.4	46.9	62.9	17.1	-448.1	28.3
MCCFormer-S	17.2	16.2	37.7	37.7	12.5	-466.4	21.3
MCCFormer-D	19.1	19.1	39.8	28.3	13.8	-476.3	24.7
VARD	24.6	22.6	40.7	42.8	14.9	-482.0	28.1
Diffusion-RSCC (Ours)	**33.3**	**27.4**	**56.5**	**88.7**	**20.4**	**-444.1**	**30.6**

LEVIR-CC and DUBAI-CC, respectively. We utilize the AdamW optimizer with an initial learning rate of $1e^{-4}$.

C. Quantitative Results

We compare the proposed method Diffuison-RSCC with previous state-of-the-art methods on the LEVIR-CC dataset and the DUBAI-CC dataset, including DUDA [44], MCCFormers-S [50], MCCFormers-D [50], RSICCformer [6], and VARD [51]. These methods are constructed on the encoder–decoder framework, in which the encoder extracts high-level image features and the decoder utilizes the LSTM-based or Transformer-based blocks to generate predicted captions. All the methods are described in detail as follows:

1) DUDA: Park et al. [44] first exploited the CC problem with data not well-aligned. There exists illumination and viewpoint change distractors in the proposed dataset CLEVR-Change by Park et al. [44] and these distractors decrease the prediction accuracy. The DUDA model consists of a dual attention module for feature extraction and mitigating the impact of distractors, and an LSTM-based dynamic speaker module for caption generation.

2) MCCFomers-S: To improve the process of caption encoding, MCCFormers-S [50] uses a Transformer caption decoder. And for the robustness to localize multiple changes, Qiu et al. [50] devised two Transformer-based encoding approaches to extract high-level image features. The first encoding approach MCCFormers-S directly feeds the feature from ResNet to the Transformer-based encoder.

3) MCCFomers-D: MCCFormers-D [50] uses the second image encoding approach devised by Qiu et al. [50] which sets a Siamese Transformers with cross-attention mechanism for feature extraction. Then, the Transformer decoder uses obtained image features to generate captions.

4) RSICCformer: RSICCformer [6] is first designed for RSI pairs. It uses a Siamese cross-attention module with different features to improve the feature discrimination capacity for changes. Also, multiple Transformer decoder layers are stacked to progressively utilize the image feature from the ResNet block to recognize changes of interest and generate captions.

5) VARD: For further mitigating the impact of distractors, VARD [51] sets a projection-based encoder including: position-embedded representation learning, unchanged representation disentanglement, and neighbor-aware position encoding. VARD successfully reduces the unchanged objects for change localization. The extracted features are then fed into the LSTM or Transformer caption generator.

We design a model structure that is completely different from previous work, reducing reliance on the encoder–decoder framework. We no longer use LSTM or Transformer for caption generation but instead rely on a probabilistic diffusion mechanism. Tables I and II summarize the quantitative results of our model and all the baselines. On the LEVIR-CC dataset, our model outperforms all tested baseline models in most metrics. On the DUBAI-CC dataset, our model outperforms all tested baselines models in all metrics. This demonstrates the generalizability of our Diffusion-RSCC. Although in Table I, the metric CIDEr column, our model is not the top performer, the difference compared with the best baseline is marginal, further validating the effectiveness of our approach.

The Diffusion-RSCC model performs best on traditional metrics, showcasing the results' accuracy and similarity with ground truth, while it also excels in the newly proposed metrics. As shown in Fig. 6, our method demonstrates its capability to maintain caption accuracy while enhancing diversity.

D. Ablation Studies

We performed two groups of additional ablation studies to validate the effectiveness of each proposed module in Diffusion-RSCC. The intricate interplay between feature extraction, cross-modal fusion, and conditional estimation modules is pivotal for accurate CC in RSIs. To systematically dissect the contributions of each component, we conducted comprehensive ablation studies focusing on architectural variations and self-attention configurations. We trained the models

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Fig. 6. Quantitative examples on the test split of datasets. (Top) Examples from the LEVIR-CC dataset. (Bottom) Examples from the DUBAI-CC dataset. We report the captions obtained by our method and all baselines along with the GT. Correct sequences are in green and incorrect parts of the captions are in red.

on the same LEVIR-CC dataset and DUBAI-CC dataset and evaluated the model performance from two aspects: 1) models with different substructures and 2) SSA with different numbers of self-attention blocks.

1) Models With Different Substructures: Tables III and IV, and Fig. 7 show the results from the first ablation study of our model on LEVIR-CC dataset and DUBAI-CC dataset. We devised different model substructures to evaluate the impact of individual modules on performance. Specifically, we explored different combinations involving ResNet101, CMF, and SSA as the feature extractor, cross-modal data mapping module, and conditional mean value estimation module.

a) Effects of the SSA module: To verify the effectiveness of the proposed SSA, we devised two methods: ResNet-CMF and ResNet-CMF-BERTEncoder. ResNet-CMF method discards the SSA module, while ResNet-CMF-BERTEncoder replaces the SSA module with the common language encoding method BERTEncoder [18] in the reverse

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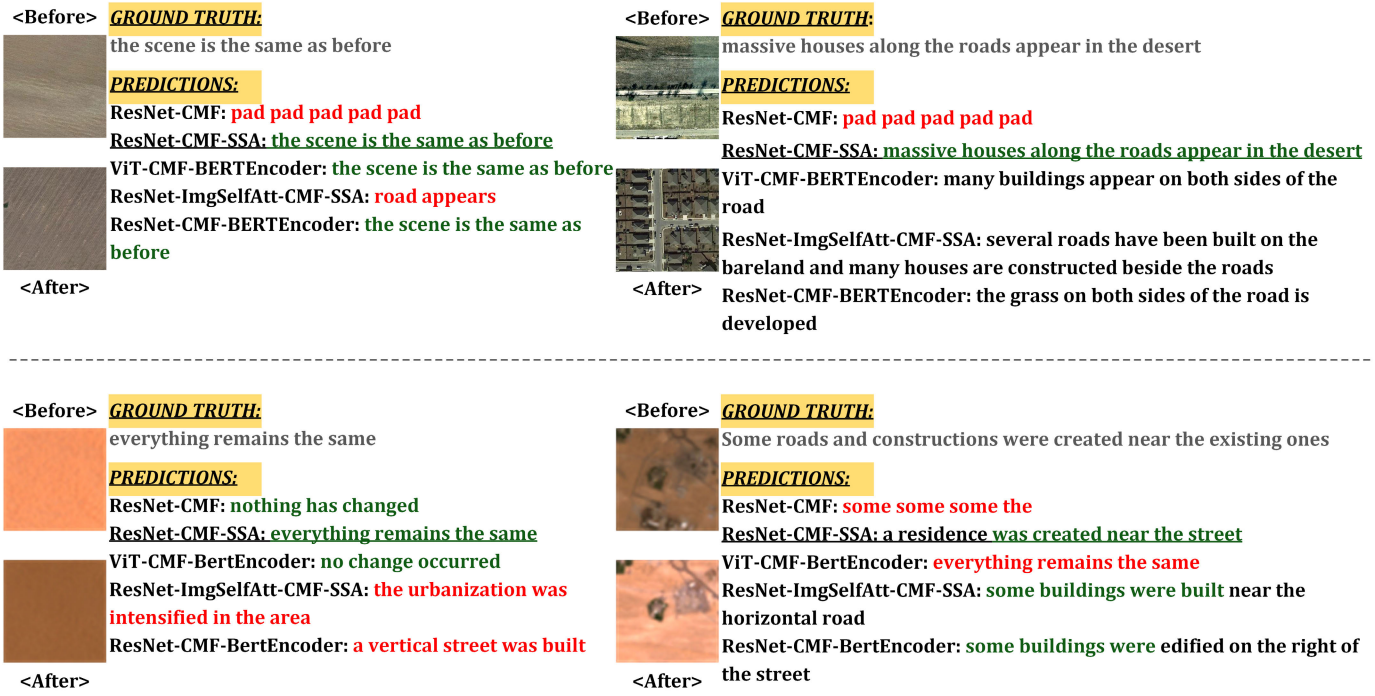


Fig. 7. Examples of the ablation study with different substructures. (Top) Examples from the LEVIR-CC dataset. (Bottom) Examples from the DUBAI-CC dataset. We report the captions of our method and baselines along with the GT. Correct sequences are in green and incorrect parts of the captions are in red.

TABLE III

ABLATION STUDY ON DIFFUSION-RSCC SUBSTRUCTURES. EXPERIMENTS ON LEVIR-CC DATASET. OUR PROPOSED MODEL OUTPERFORMS ALL BASELINES ON ALL THE METRICS. THE NUMBERS ARE IN %. THE BEST RESULTS ARE IN BOLD

Method	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE	BARTScore	MoverScore
ResNet-CMF	3.4	2.5	7.9	2.9	1.3	-571.0	3.6
ViT-CMF-BertEncoder	37.5	26.7	52.8	78.32	18.4	-454.0	25.9
ResNet-CMF-BertEncoder	55.3	36.9	70.1	120.4	26.6	**-417.7**	32.2
ResNet-ImgSelfAtt-CMF-SSA	37.9	29.2	56.6	87.6	20.7	-444.5	27.2
ResNet-CMF-SSA(Ours)	**60.9**	**37.8**	**71.8**	**125.6**	**26.7**	**-417.7**	**32.4**

TABLE IV

ABLATION STUDY ON DIFFUSION-RSCC SUBSTRUCTURES. EXPERIMENTS ON DUBAI-CC DATASET. OUR PROPOSED MODEL OUTPERFORMS ALL BASELINES ON MOST OF THE METRICS. THE NUMBERS ARE IN %. THE BEST RESULTS ARE IN BOLD

Method	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE	BARTScore	MoverScore
ResNet-CMF	0.0	7.0	22.2	17.7	5.2	-536.7	12.1
ViT-CMF-BertEncoder	29.2	25.2	49.0	67.2	19.8	-449.3	28.6
ResNet-CMF-BertEncoder	32.8	26.	54.9	80.8	19.1	**-442.8**	**30.9**
ResNet-ImgSelfAtt-CMF-SSA	22.3	23.2	47.0	57.9	18.7	-456.4	27.4
ResNet-CMF-SSA(Ours)	**33.3**	**27.4**	**56.5**	**88.7**	**20.4**	-444.1	30.6

process of mean value estimation. Tables III and IV shows the results from Res-Net-CMF, ResNet-CMF-BERTEncoder, and ResNet-CMF-SSA. Compared with ResNet-CMF-SSA, ResNet-CMF performed poorly on all metrics, indicating that it was unable to generate natural language correctly and demonstrating the capability of the SSA module in change caption generation. The results of ResNet-CMF-BERTEncoder outperform all other models except for ResNet-CMF-SSA. It shows the capability of the BERTEncoder in mean value estimation. However, our ResNet-CMF-SSA (Diffusion-RSCC) achieves the best performance across all evaluations on the LEVIR-CC dataset and most evaluations on the DUBAI-CC dataset, underscoring the great effectiveness of the SSA module.

b) Effects of the ResNet module: Residual Network (ResNet101) [53] is a CNN with a depth of 152 layers that democratized the concepts of residual learning and skip connections. It has helped extracting object-level features in previous CC tasks [63]. To further demonstrate the effectiveness of ResNet, we compare the ResNet module with the mainstream visual model Vision Transformer (ViT) [64]. As shown in Tables III and IV, the ResNet-CMF-BERTEncoder significantly outperformed ViT-CMF-BERTEncoder across all metrics except SPICE, indicating the superior effectiveness of the ResNet module in single-temporal image processing.

c) Effects of the CMF module: To test the effectiveness and robustness of the proposed CMF, We devised an ImgSelfAtt block, inspired by RSICCformer. In ResNet-ImgSelfAtt-CMF-SSA, the ImgSelfAtt is set for capturing dependencies and relationships within single-temporal image feature $\varphi(I_{\text{bef}})$ and $\varphi(I_{\text{aft}})$. However, Tables III and IV show that the evaluation results of ResNet-ImgSelfAtt-CMF-SSA

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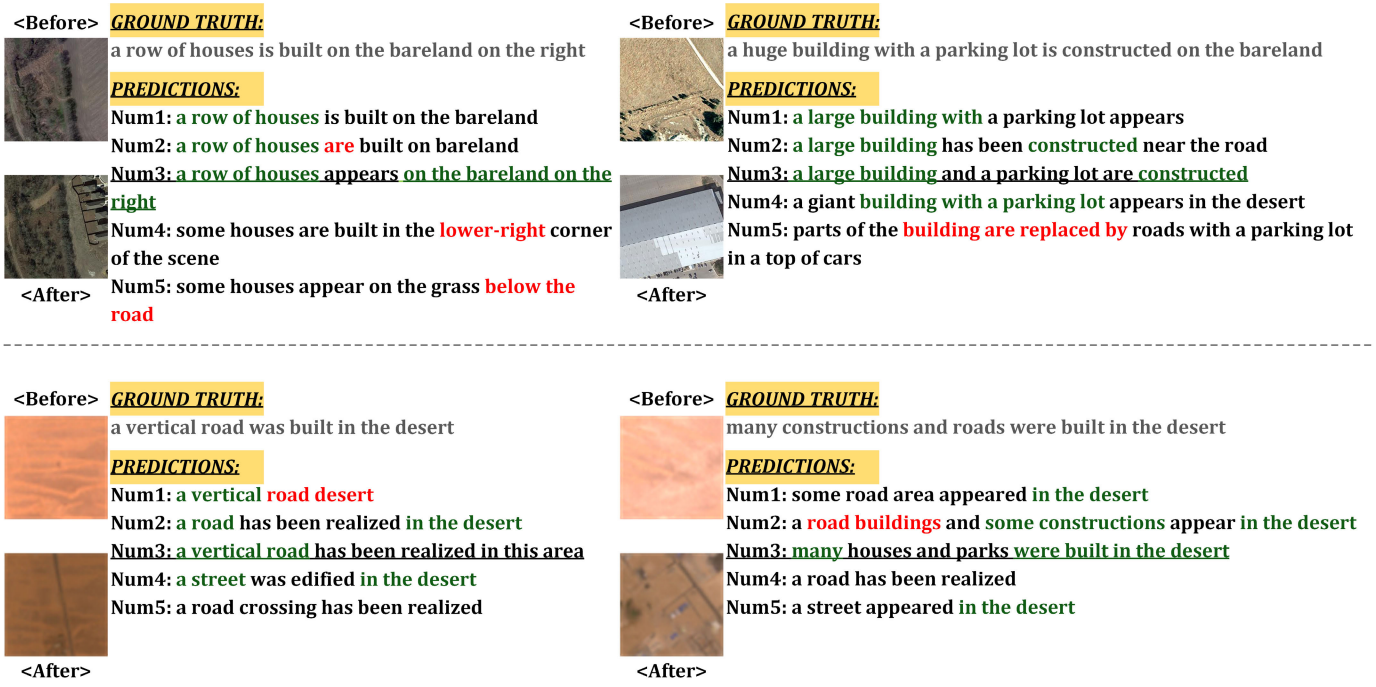


Fig. 8. Examples of the ablation study about SSA with different numbers of self-attention blocks. (Top) Examples from the LEVIR-CC dataset. (Bottom) Examples from the DUBAI-CC dataset. We report the captions of our method and 1–5 block baselines along with the GT. Correct sequences are in green and incorrect parts of the captions are in red.

TABLE V

ABLATION STUDY ON SSA WITH DIFFERENT NUMBERS OF SELF-ATTENTION BLOCKS. EXPERIMENTS ARE TAKEN ON THE LEVIR-CC DATASET. OUR PROPOSED MODEL OUTPERFORMS ALL BASELINES ON ALL THE METRICS. THE NUMBERS ARE IN %. THE BEST RESULTS ARE IN BOLD

Method	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE	BARTScore	MoverScore
1	50.2	19.2	56.5	96.3	21.0	-452.1	26.1
2	57.4	35.7	68.2	119.7	25.7	-423.7	31.1
3	**60.9**	**37.8**	**71.5**	**125.6**	**26.7**	**-417.7**	**32.4**
4	55.8	35.5	68.0	117.9	25.6	-424.9	31.0
5	50.9	34.7	66.9	113.1	25.4	-424.6	30.5

TABLE VI

ABLATION STUDY ON SSA WITH DIFFERENT NUMBERS OF SELF-ATTENTION BLOCKS. EXPERIMENTS ARE TAKEN ON DUBAI-CC DATASET. OUR PROPOSED MODEL OUTPERFORMS ALL BASELINES ON MOST OF THE METRICS. THE NUMBERS ARE IN %. THE BEST RESULTS ARE IN BOLD

Method	BLEU-4	METEOR	ROUGE-L	CIDEr	SPICE	BARTScore	MoverScore
1	23.2	18.4	41.1	60.5	15.7	-480.9	22.5
2	27.8	24.8	51.0	73.0	17.4	-451.6	22.5
3	**33.3**	**27.4**	**56.5**	**88.7**	**20.4**	-444.1	30.6
4	29.7	26.5	55.2	84.2	20.2	**-443.7**	**30.9**
5	29.7	26.3	54.1	82.2	18.7	-447.0	30.3

are significantly lower than those of the ResNet-CMF-SSA. This suggests that the ImgSelfAtt block might lead to overfitting within single-temporal image features, reducing the accuracy of caption generation in bitemporal scenarios. It also demonstrates that CMF is highly efficient in mapping between bitemporal image features and text representation.

2) SSA With Different Numbers of Self-Attention Blocks: For the ablation study on the number of self-attention blocks in SSA, we evaluated the effectiveness of different configurations by setting the number of self-attention blocks from 1 to 5. As shown in Table V, performance across all metrics significantly improves as the number of self-attention blocks increases, peaking when there are three blocks. When the number of blocks exceeds three, performance declines as additional blocks are added. In Table VI, the trend is similar. However, for the BARTScore and MoverScore metrics, the model reached its peak when the number is 4.

As shown in Fig. 8, before the number of blocks reaches 3, the caption generation model is improved in both grammar learning and accuracy under semantic changes. Beyond three blocks, the additional block can enhance the learning of language rules and the flexibility of caption generation, but it negatively impacts the accuracy of caption generation. While deeper architectures may improve linguistic richness, they introduce noise that hinders precise semantic alignment. In Fig. 8, due to the relatively small sample size and simpler text structure in the DUBAI-CC dataset, this trend of increasing text complexity with more blocks is not observed. Through extensive experimentation and careful analysis, we ultimately settled on using the SSA module with three self-attention blocks in our proposed Diffusion-RSCC.

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V. Conclusion

In this article, we propose a novel diffusion model (Diffusion-RSCC) for RSICC to learn the probabilistic distribution of the inputs through the forward noising process and reverse denoising process. In the reverse process, a condition denoiser is designed to estimate the mean value conditioned on the bitemporal image pairs. The image feature extractor ResNet101 module and the newly proposed CMF and SSA in the condition denoiser were evaluated on the LEVIR-CC dataset and DUBAI-CC dataset. Through extensive experiments and analysis, we demonstrate the effectiveness of Diffusion-RSCC and each of its individual modules. Diffusion-RSCC achieves superior performance compared with previous CC methods.

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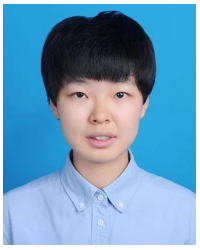
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